
EDUCATION

University of Oxford*PhD in Statistical Machine Learning*

Oct 2024 – Apr 2028 (expected)

- Research Area: Generative Modelling, Sampling, Monte Carlo, Optimisation
- Supervisors: Yee Whye Teh & George Deligiannidis
- Scholarships: Google DeepMind PhD Scholarship

University of New South Wales*Master of Mathematics*

Jun 2020 – Dec 2023

Master of Information Technology

Jul 2017 – Jan 2020

*Bachelor of Commerce/Science*Feb 2011 – Dec 2015

PUBLICATIONS & PREPRINTS

First or shared-first author:

- *Metropolis-Adjusted Diffusion Models*, **K. H. Lam***, T. Farghly*, C. Williams*, J. Yang, Y. W. Teh, A. Doucet, arXiv (Under Review)
- *Rao-Blackwellised Reparameterisation Gradients*, **K. H. Lam**, T. D. Bui, G. Deligiannidis, Y. W. Teh, Conference on Neural Information Processing Systems 2025, NeurIPS 2025
- *LegendreTron: Uprising Proper Multiclass Loss Learning*, **K. H. Lam**, C. Walder, S. Penev, R. Nock, International Conference on Machine Learning 2023, ICML 2023

Co-author:

- *Permutation-Invariant Spectral Learning via Dyson Diffusion*, T. Schwarz, C. Dieball, C. Kogler, **K. Lam**, R. Lambiotte, A. Doucet, A. Godec, G. Deligiannidis, arXiv (Under Review)
 - *Regularized Black-box Optimization Algorithms for Least-squares Problems*, Y. Liu, **K. H. Lam**, L. Roberts, IMA Journal of Numerical Analysis, IMAJNA 2025
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RESEARCH EXPERIENCE

Microsoft – Research Intern

Apr 2026 – Jul 2026

- Research on Image Editing with Diffusion and Flow models

Australian National University – Research Intern / Research Assistant

Nov 2022 – Oct 2024

- Research on Variational Inference for Discrete Neural Networks

University of Technology Sydney – Research Assistant

Nov 2023 – Apr 2024

- Research on Structure Learning for Bayesian Networks and LLMs

Australian National University – Research Intern

Nov 2023 – Jan 2024

- Research on Control Theory and Differential Geometry

Australian National University – Research Intern

Nov 2021 – Jan 2022

- Research on Regularised Black-box Optimisation algorithms

Department of Defence (DSTG) – Research Cadet

May 2021 – Nov 2021

- Research on Adversarial Machine Learning

University of Sydney – Research Intern / Research Engineer

Dec 2020 – May 2021

- Research on Manifold MCMC algorithms
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SKILLS

- Programming – Python, SQL, R, Matlab, C, C++
- Python Libraries – PyTorch, JAX, Tensorflow, XGBoost, LightGBM, scikit-learn, pandas, NumPy, SciPy
- Tooling – Linux, Git, Kubernetes, Docker, Docker Compose
- Computing Services – Databricks, Spark, Azure, AWS, GCP, HPC, Slurm

TEACHING

SC5 Advanced Simulation Methods – <i>Teaching Assistant</i>	Jan 2026 – Mar 2026
SB2.2 Statistical Machine Learning – <i>Teaching Assistant</i>	Jan 2025 – Mar 2025
COMP9444 Neural Nets & Deep Learning – <i>Consultation Tutor</i>	Sep 2020 – Dec 2020

INDUSTRY EXPERIENCE

QBE – <i>Machine Learning Engineer</i>	Jan 2024 – Sep 2024
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- Automated **Docker** workflows on **JFrog Artifactory** using **Azure pipelines** as CI/CD
- Deployed a database backup service using **Docker Compose** and **CRON**

Hivery – <i>Machine Learning Engineer</i>	Aug 2022 – Jan 2024
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- Led the adoption of **MLOps** practices by using **Azure Databricks** and **MLflow** to save and track models
- Automated building and storage of model container images by using **Docker** and **Gitlab pipelines** as CI/CD
- Implemented and deployed demand transfer models for retail assortment

Woolworths (WooliesX) – <i>Senior Data Scientist</i>	Jul 2018 – Oct 2019
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- Implemented **GLM** and **GBM** models for loyalty programs including **Personalisation** and **Customer Lifetime Value**
- Illustrated effectiveness of ML-driven marketing through a **randomised controlled trial** and **A/B testing**

Quantium – <i>Data Analyst</i>	Nov 2017 – May 2018
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- Identified home delivery expansion opportunities for a national retailer by using analytics generated from **PySpark**

Commonwealth Bank – <i>Quantitative Analyst</i>	Jan 2017 – Oct 2017
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- Developed **IRB models** for regulatory capital by modelling **probabilities of default** against risk factors with **GLMs**

AMP – <i>Financial Risk Analyst</i>	Jan 2016 – Dec 2016
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PROFESSIONAL ACCREDITATIONS

Global Association of Risk Professionals – <i>Financial Risk Manager</i>	Sep 2021 – Present
Statistical Society of Australia – <i>Graduate Statistician</i>	Nov 2020 – Dec 2023
CFA Institute – <i>Chartered Financial Analyst</i>	Sep 2020 – June 2021